

Thermo 2020-2021

Exercice 1a/ $k=0$; la transf est isothermeb) Equation d'Etat

$$PV = nRT = \frac{m}{M_{\text{air}}} RT \Rightarrow PV = \frac{R}{M_{\text{air}}} T$$

c) Loi de la statique des fluides

$$dP = -\rho g dz \Rightarrow \frac{dP}{dz} = -\rho g \quad \text{or} \quad \rho = \frac{m}{V} = \frac{1}{v}$$

$$\frac{dP}{dz} = -\frac{g}{v} \quad \text{ou} \quad dP = -g \frac{1}{v} dz$$

d) Expression de $\frac{dT}{dz} = \alpha$

$$\text{on a: } P^{-k} T = P_0^{-k} T_0 \Rightarrow T = T_0 \cdot \left(\frac{P}{P_0}\right)^k$$

$$\frac{dT}{dz} = T_0 \left(\frac{1}{P_0}\right)^k \cdot k P^{k-1} \frac{dP}{dz} = k T_0 \left(\frac{P}{P_0}\right)^{\frac{1}{k}} \frac{1}{P} \frac{dP}{dz}$$

$$= \frac{k}{P} \cdot T_0 \left(\frac{P}{P_0}\right)^k \frac{dP}{dz} = \frac{k T}{P} \frac{dP}{dz} \quad \text{or} \quad \frac{dP}{dz} = -\frac{g}{v}$$

$$\frac{dT}{dz} = \frac{k}{P} T \left(-\frac{g}{v} \right) = -kg \cdot \frac{T}{Pv} \quad \text{or } Pv = \frac{R}{M}$$

$$= -kg \cdot \frac{M}{R}$$

$$\Rightarrow \frac{dT}{dz} = - \frac{kMg}{R} = \alpha = \frac{-0,15 \times 29 \times 10^{-3} \times 10}{8,32}$$

$$\alpha = -5,23 \times 10^{-3} \text{ K m}^{-1}$$

e) Variation de $T(z)$

$$\frac{dT}{dz} = \alpha \Rightarrow T = T_0 + \alpha z \quad (z=0, T=T_0)$$

f) Expression de $\frac{V(z)}{V_0}$

$$\text{ona: } \left. \begin{array}{l} PV = nRT \\ P_0 V_0 = nRT_0 \end{array} \right\} \frac{PV}{P_0 V_0} = \frac{T}{T_0} \Rightarrow \frac{V}{V_0} = \frac{T}{T_0} \cdot \frac{P_0}{P}$$

$$\frac{V}{V_0} = \frac{T}{T_0} \cdot \frac{P_0}{P} \quad \text{or} \quad \frac{P_0}{P} = \left(\frac{T}{T_0} \right)^{-\frac{1}{k}}$$

$$\frac{V}{V_0} = \frac{T}{T_0} \cdot \left(\frac{T}{T_0} \right)^{-\frac{1}{k}} = \left(\frac{T}{T_0} \right)^{1 - \frac{1}{k}} = \left(\frac{T}{T_0} \right)^{\frac{k-1}{k}}$$

$$\text{or } T = T_0 + \alpha z$$

$$\Rightarrow \frac{V(z)}{V_0} = \left[\frac{T_0 + \alpha z}{T_0} \right]^{\frac{k-1}{k}} = \left[1 + \frac{\alpha}{T_0} z \right]^{\frac{k-1}{k}}$$

Exercice 2

(2)

1° Calcul des grandeurs à la sortie

$$\Delta m (\Delta h + \Delta e_c + \cancel{\Delta e_p}) = \cancel{\dot{Q}_u} + \cancel{\dot{Q}_{th}}$$

$$\Delta h + \Delta e_c = 0 \Rightarrow h_s - h_e = \frac{1}{2} (C_1^2 - C_2^2)$$

$$h_s = h_e + \frac{1}{2} (C_1^2 - C_2^2) \text{ or } h_e = 2851 \text{ kJ/kg}$$

$$h_s = 2851 + \frac{1}{2} (50^2 - 600^2) \times \left(\frac{1}{1000}\right) \text{ - Conversion en kJ}$$

$$h_s = 2672,25 \text{ kJ/kg} < h_g(1,56) = 2693 : \text{Vapeur humide}$$

$$T = 111,4^\circ\text{C}$$

$$x = \frac{2672,25 - 467}{2693 - 467} = 0,99$$

$$s = x(s_g - s_f) + s_f = 0,99 \times (7,223 - 1,434) + 1,434 \\ = 7,16511 \text{ kJ/kgK}$$

$$v = x(v_g - v_f) + v_f = x v_g = 1,1474 \text{ m}^3/\text{kg}$$

2° Calcul de la vitesse maximale

Écoulement réversible $\Rightarrow$ Adiabatique réversible d'où
isentropique $\Rightarrow s_1 = s_2 = 6,968 \text{ kJ/kgK}$

$$x = \frac{6,968 - 1,434}{7,223 - 1,434} = 0,9556$$

$$h_2 = x(h_g - h_f) + h_f = 0,9556 \times (2693 - 467) + 467$$

$$h_2 = 2594,165 \text{ kJ/kg}$$

$$\text{or } h_2 - h_1 = \frac{1}{2} (C_1^2 - C_2^2)$$

$$\Rightarrow 2(h_1 - h_2) = \frac{1}{2} C_2^2 - C_1^2 \Rightarrow C_2^2 = 2(h_1 - h_2) + C_1^2$$

$$C_2 = \left[2(h_1 - h_2) + C_1^2 \right]^{\frac{1}{2}} = \left[2(2851 - 2594,16) + 450^2 \right]^{\frac{1}{2}}$$

$$V_{\max} = 718,4567 \text{ m/s}$$

Conversion
en J.

Exercice 3

$$1^\circ / \begin{aligned} h_1 &= h_2 = h_f(T_1) = 448 \text{ kJ/kg} \\ x_2 &= \frac{h - h_f(T_2)}{h_g - h_f(T_2)} = \frac{448 - 387}{556 - 387} = 0,36 : \text{Vapeur humide.} \end{aligned}$$

$$\Delta S = S_2 - S_1 \quad S_1 = s_f(T_1) = 4,286 \text{ kJ/kg K.}$$

$$\begin{aligned} S_2 &= x(s_g - s_f) + s_f(T_2) = 0,36 \cdot (4,776 - 4,061) + 4,061 \\ &= 4,3184 \text{ kJ/kg K} \end{aligned}$$

$$\Delta S = S_2 - S_1 = 4,3184 - 4,286 = 0,0324 \text{ kJ/kg K}$$

$$\Delta S = 32,4 \text{ J/kg K} > 0 \text{ Transf. irréversible.}$$

$$2^\circ / \begin{aligned} \Delta h &= h_2 - h_1 = 0 = C_L(T_2 - T_1) + L_v(x_2 - 0) \\ &= C_L(T_2 - T_1) + x_2 L_v = 0 \end{aligned}$$

$$\Rightarrow x_2 = \frac{C_L}{L_v} (T_1 - T_2) = \frac{992}{168} (303 - 237) \quad (3)$$

$$x_2 = 0,3614 \simeq 0,36$$

$$\bullet \Delta S = S_2 - S_1 = C_L \ln\left(\frac{T_2}{T_1}\right) + \frac{x L_v}{T_2}$$

$$\Delta S = 0,92 \ln\left(\frac{237}{303}\right) + \frac{0,36 \times 168}{237} = 0,029171$$

$$\Delta S = 29,171 \text{ J/Kg K}$$

les résultats sont du même ordre dans les 2 cas.

Exercice 1a- Calcul de T_2

$$\text{on a: } \dot{m} \left[h_2 - h_1 + \frac{1}{2}(C_2^2 - C_1^2) + g(z_2 - z_1) \right] = \dot{Q}_u + \dot{Q}_h = 0$$

$$\Rightarrow \dot{m}(h_2 - h_1) = C_p(T_2 - T_1) \times \dot{m} = \dot{Q}_u$$

$$\Rightarrow T_2 = \frac{\dot{Q}_u}{\dot{m} C_p} + T_1 = 530,769 \text{ K} = \frac{1,5 \cdot 10^3}{6,5 \cdot 10^3 \times 10^3} + 300 \text{ K}$$

b) Calcul de ΔS

$$dS = C_p \frac{dT}{T} + \frac{h}{T} dP = C_p \frac{dT}{T} - \frac{v}{T} dP = C_p \frac{dT}{T} - \frac{R}{M} \frac{dP}{P}$$

$$= C_p \frac{dT}{T} - r \frac{dP}{P} \Rightarrow \Delta S = C_p \ln \frac{T_2}{T_1} - r \ln \left(\frac{P_2}{P_1} \right)$$

$$\text{or } C_p = \frac{r r}{r-1} \Rightarrow r = \frac{(r-1) C_p}{r}$$

$$\Delta S = C_p \ln \frac{T_2}{T_1} - \frac{r-1}{r} C_p \ln \frac{P_2}{P_1}$$

$$\Delta S = 1 \ln \left(\frac{530,76}{300} \right) - \frac{0,4}{1,4} \times 1 \ln(6) = 58,568 \text{ J/Kg.K.}$$

$$\dot{m} \Delta S = 6,5 \cdot 10^3 \times 58,568 = 0,38 \text{ J/K.S.} > 0$$

$\Rightarrow$ Transformation irréversible (Adiabatique irréversible)

c) Calcul du nouveau débit massique

Transf. isentropique $\Rightarrow$ Adiabatique réversible

d'où $T_2^\gamma P_2^{1-\gamma} = T_1^\gamma P_1^{1-\gamma} \Rightarrow T_2 = T_1 \cdot \left(\frac{P_1}{P_2}\right)^{\frac{1-\gamma}{\gamma}}$

$$T_2 = 300 \left(\frac{1}{6}\right)^{-\frac{0,4}{1,4}} = 500,55 \text{ K.}$$

$$Q_u = D_m c_p (T_2 - T_1) \Rightarrow D_m = \frac{Q_u}{c_p (T_2 - T_1)}$$

$$D_m = \frac{1,5 \cdot 10^3}{10^3 (500,55 - 300)} = 7,479 \cdot 10^{-3}$$

$$D_m = 7,479 \cdot 10^{-3} \text{ kg/s} = 7,479 \text{ g/s}$$

Exercice 2

- 1°/ a. Etat I : Vapeur (vapeur sèche / surchauffée)
" A : Mélange liq+vap. saturé en vapeur ou vapeur saturé
B : Mélange liq+vap
D : Mélange liquide + vap saturé en liquide ou liquide saturé
F : liquide

b/ Titre massique en B

$$x_B = \frac{D_B}{D_A} = \frac{0,3 - 0,6}{0,3 - 1} = 0,43$$

2°/ a. Calcul des Transferts thermiques I à F

$$Q = Q_{IA} + Q_{AB}$$

$$\Delta U_{IA} = Q_{IA} + W_{IA} = 0 \Rightarrow Q_{IA} = -W_{IA}$$

②

$$Q_{IA} = \int_I^A P dv = n_0 R T \ln \frac{V_A}{V_I} = -3,616 \text{ J}$$

$$Q_{AB} = -n_B L_{\text{vap}} \text{ (changement d'état)}$$

$$= -(1-x_B) n_0 L_{\text{vap}} = -8,717 \text{ J}$$

$$Q = -3,616 - 8,717 = -12,333 \text{ J}$$

$$W_{IA} + W_{AB}$$

$$W_{IA} = -Q_{IA} = 3,616 \text{ J}$$

$$W_{AB} = - \int_A^B P dv = -P(V_B - V_A) = 1,14 \text{ J}$$

$$W = 3,616 + 1,14 = 4,75 \text{ J}$$

b) calcul ΔU_{IAB}

$$\Delta U_{IAB} = W + Q = -7,55 \text{ J}$$

c) calcul de ΔS

$I \rightarrow A \rightarrow B$: Transf. réversible isotherme

$$\Delta S = S_e = \frac{Q}{T} = \left(n_0 R \ln \frac{V_A}{V_I} \right) - (1-x_B) n_0 \frac{L_{\text{vap}}}{T}$$

$$\Delta S = -4,10^{-2} \text{ J K}^{-1}$$

$$\Delta S_{IAB} = S_e \text{ donc } S_e = 0 \text{ J K}^{-1}$$